

NEWPORT NEWS DOMINION TERM, VA, USA

WMO: 997782

Lat: 36.962N

Lon: 76.424W

Elev: 7

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.2	(c) 25.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 30.1	(k) 35.0	(l) 27.0	(m) 41.5	(n) 10.5	(o) 0	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	89.5	N/A	87.5	N/A	85.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.8	(b) 22.3	(c) 19.2	(e) DB	(f) 18.7	(g) 94.3	(h) 4.7	(i) 3.0	(j) 15.3	(k) 96.5	(l) 12.5	(m) 98.2	(n) 9.9	(o) 99.9	(p) 6.4	(q) 102.1	(r) N/A
(4) 25.8	(5) 22.3	(6) 19.2	(7) DB	(8) 18.7	(9) 94.3	(10) 4.7	(11) 3.0	(12) 15.3	(13) 96.5	(14) 12.5	(15) 98.2	(16) 9.9	(17) 99.9	(18) 6.4	(19) 102.1	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	40.3	42.5	48.5	59.9	68.5	76.7	80.2	78.5	73.8	64.4	52.2	47.1
	DBStd	15.44	8.93	8.19	8.18	7.10	6.10	4.74	3.53	3.19	4.64	6.61	6.71	7.97
	HDD50	896	319	234	129	8	0	0	0	0	0	2	54	150
	HDD65	3158	765	629	511	184	37	1	0	0	1	89	385	556
	CDD50	4966	20	25	84	303	572	802	935	883	715	448	119	60
	CDD65	1753	0	0	1	29	144	353	470	418	266	70	1	1
	CDH74	12513	0	0	1	71	564	2648	4403	3297	1367	162	0	0
	CDH80	2996	0	0	0	4	86	695	1322	682	196	11	0	0
(14) Wind	WSAvg	9.1	9.6	9.8	9.3	9.8	9.4	9.1	9.4	8.1	8.2	9.0	8.5	9.1
(15) Precipitation	PrecAvg	47.5	3.2	3.0	3.7	3.3	3.5	4.2	5.5	5.3	5.4	3.9	3.1	3.3
	PrecMax	60.4	5.9	8.6	9.9	6.0	5.3	10.4	9.4	13.6	17.5	8.1	11.0	6.8
	PrecMin	30.8	1.1	0.9	0.6	0.6	1.3	1.7	1.3	1.6	0.7	0.3	0.2	1.3
	PrecStd	8.1	1.3	1.6	1.7	1.3	1.0	2.0	2.0	2.9	3.9	2.1	2.3	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.0	66.5	71.4	79.2	86.8	92.0	93.8	89.9	86.9	81.4	69.1	68.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	60.9	62.0	66.6	76.0	82.4	88.5	90.4	87.5	84.4	77.7	66.2	65.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	57.1	58.7	63.8	73.1	79.8	86.2	88.2	85.8	82.3	75.5	64.1	61.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	53.7	54.9	60.8	70.2	77.3	84.2	86.2	83.7	80.3	73.4	62.0	58.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.8	11.8	12.1	11.2	10.4	10.0	8.7	8.5	9.6	10.2	10.5
		MCDBR	15.9	16.5	16.1	15.5	13.9	13.4	12.6	11.6	10.9	10.9	11.8	13.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.307	0.320	0.347	0.392	0.435	0.486	0.513	0.507	0.408	0.356	0.324	0.312
	taud		2.538	2.498	2.453	2.340	2.249	2.159	2.097	2.100	2.369	2.509	2.547	2.558
	Ebn at Noon		282	291	292	282	270	256	247	246	268	274	273	273
	Edh at Noon		25	29	33	39	44	48	50	49	36	28	24	23
(44) All-Sky Solar Radiation	RadAvg		747	986	1311	1669	1878	2017	1920	1731	1408	1121	843	665
	RadStd		61	107	104	122	163	130	94	95	116	130	69	53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+193

Nomenclature: See separate page